

Martin de Juan

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EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Arts in Computer Science and Math

Sep. 2022 – May 2026

- Courses: Computation, Programs, and Programming — 97%; CPSC V 121 - Models of Computation — 91%; Software Construction — 96%; Introduction to Computer Systems — 91%; Basic Algorithms and Data Structures — 90%; Intermediate Algorithm Design and Analysis — 96%

EXPERIENCE

Undergraduate Teaching Assistant

Sep. 2023 – Present

University of British Columbia CPSC 110/107, Computation, Programs, and Programming

Vancouver, BC

- Taught proper coding conventions, basic recursion, and fundamental programming constructs.
- Guided students through generic searches, graphs, and data structures including arrays and binary trees.
- Held office hours, graded homework, and invigilated exams.

Undergraduate Teaching Assistant

July 2025 – Aug. 2025

University of British Columbia CPSC 320, Intermediate Algorithm Design and Analysis

Vancouver, BC

- Taught core algorithms including basic algorithmic techniques, recursive dynamic programming/memoization, divide and conquer, and NP-hard proofs.
- Led tutorials solving algorithmic problems and provided step-by-step explanations.
- Graded weekly assignments and held office hours.

PROJECTS

Course Insights Dashboard | *TypeScript, Node.js, Express, React, HTML/CSS, JSON*

Jan 2025 – April 2025

- Engineered a full-stack web application for analyzing university course data from uploaded ZIP datasets.
- Implemented a backend API using Node.js and Express to manage dataset uploads, extraction, and storage, supporting operations to add, remove, and list datasets dynamically.
- Developed a custom JSON query parser to filter, sort, and aggregate course information—enabling queries such as course trends over years, instructor averages, and departmental performance.
- Built an interactive React frontend displaying insights including top instructors, average grades by course and department, and instructor-specific course histories.
- Designed persistent state management to retain datasets and user preferences (including dark mode) across refreshes and browser sessions.
- Focused on clean architecture separating frontend and backend concerns while ensuring fast query responses and intuitive data visualization.

Store Inventory Manager | *Java, Swing*

Sep. 2023 – Dec. 2023

- Developed an object-oriented Java application to help small stores manage product inventory, prices, and quantities.
- Implemented core features for adding, updating, and removing products, processing incoming shipments, and handling customer sales.
- Designed logic to calculate total sale price dynamically and adjust stock levels after each transaction.
- Built a simple graphical user interface using Swing for intuitive interaction with the system, including input forms and sales displays.
- Emphasized clean code structure using classes and collections (ArrayLists, HashMaps) to store and manage product data efficiently.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, HTML/CSS, R, Assembly

Frameworks: React, Node.js, JUnit, Mocha, Chai.js, supertest

Developer Tools: Git, VS Code, Visual Studio, Jupyter, IntelliJ

Libraries: pandas, NumPy, Matplotlib, scikit learn